

Bias Detection and Fairness Analysis in Machine Learning Models

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Project Domain: Machine Learning / Ethical AI

Abstract

This project focuses on detecting and mitigating bias in machine learning models using fairness-aware techniques. A Random Forest classifier was trained on the Adult Census Income dataset to predict income categories. Fairness metrics such as Demographic Parity Difference and Equalized Odds Difference were used to evaluate bias associated with sensitive attributes such as gender. Bias mitigation was performed using the Exponentiated Gradient Reduction algorithm from the Fairlearn library. The project demonstrates how fairness evaluation can improve ethical AI systems while maintaining acceptable prediction accuracy.

1. Introduction

Machine learning systems are increasingly used in hiring, finance, healthcare, surveillance, and recommendation systems. However, these systems can unintentionally learn bias from historical data. This project analyzes algorithmic bias in a machine learning model and applies fairness mitigation techniques to improve ethical decision-making.

2. Objectives

- Build a machine learning model for income prediction.
- Analyze fairness across demographic groups.
- Evaluate fairness metrics before and after mitigation.
- Reduce algorithmic bias using Fairlearn.
- Generate fairness reports and visualizations.

3. Methodology

The project workflow consists of the following stages:

1. Dataset Collection and Preprocessing
2. Handling Missing Values
3. Encoding Categorical Variables
4. Splitting Dataset into Training and Testing Sets
5. Training a Random Forest Classifier
6. Evaluating Model Accuracy and Classification Metrics
7. Measuring Fairness Metrics
8. Applying Bias Mitigation using Exponentiated Gradient Reduction
9. Generating Reports and Visualizations

4. Tools and Technologies

- Python 3.11
- Pandas
- NumPy
- Scikit-learn
- Fairlearn
- Matplotlib
- Seaborn

5. Dataset Description

The Adult Census Income dataset was used in this project. The dataset contains demographic and employment-related information such as age, education, occupation, workclass, race, gender, and hours worked per week. The target variable predicts whether annual income exceeds \$50K.

6. Implementation Details

The model was implemented using a Random Forest classifier due to its strong performance on structured tabular datasets. Fairness metrics were calculated using the Fairlearn library. The model outputs were analyzed before and after mitigation to study fairness tradeoffs.

7. Results and Observations

The trained model achieved an accuracy of approximately 85.61%. Initial fairness analysis revealed measurable demographic disparity. After applying fairness mitigation techniques, demographic parity difference reduced significantly from 0.1730 to 0.0168.

Metric	Before Mitigation	After Mitigation
Accuracy	85.61%	Slight Reduction
Demographic Parity Difference	0.1730	0.0168
Equalized Odds Difference	0.0758	0.1328

8. Future Scope

- Develop a real-time fairness monitoring dashboard.
- Compare multiple machine learning algorithms.
- Integrate explainable AI techniques such as SHAP and LIME.
- Deploy the fairness-analysis model as a web application.

9. Conclusion

This project successfully demonstrated how machine learning models can contain hidden bias and how fairness-aware algorithms can reduce such disparities. The implementation highlighted the importance of ethical AI and responsible machine learning practices. The project also demonstrated practical applications of fairness metrics and bias mitigation techniques.

10. References

1. Fairlearn Documentation
2. Scikit-learn Documentation
3. UCI Adult Census Income Dataset
4. Research Papers on Algorithmic Fairness and Ethical AI